

## SYNOPSIS

At 0540 on 10 March 2012, the bulk carrier *Seagate* and the refrigerated-cargo ship *Timor Stream* collided while transiting open waters, in good conditions of visibility, 24 nm north of the Dominican Republic. There were no injuries, but both ships were badly damaged and there was some minor pollution.

*Timor Stream* left port 3 hours before the collision and was proceeding to the United Kingdom; *Seagate* was on passage to the west coast of Africa. *Seagate*'s chief officer saw *Timor Stream* but assumed it was an overtaking vessel which would keep clear of *Seagate*. The master of *Timor Stream*, who was alone on the bridge, was not keeping an effective lookout. Neither watchkeeper realised that the two vessels were on a collision course until less than a minute before the accident.

Poor watchkeeping standards, driven by complacency, led to the collision. The officer in charge of the navigational watch on both vessels failed to keep a proper lookout, did not assess the risk of, or take appropriate action to avoid collision. In summary, both officers failed to comply with some of the most fundamental elements of the International Regulations for Preventing Collisions at Sea 1972 (as amended) and the written navigational procedures issued by their respective company managers.

The managers of both vessels have taken action designed to prevent similar accidents in the future which address the safety issues identified in the MAIB's investigation. Accordingly, no recommendations have been issued with this report.

## 1.2 NARRATIVE

### 1.2.1 Events leading to the collision

On 3 March 2012 *Seagate* departed Beaumont, Texas, United States of America (USA), with a cargo of wheat for carriage to Lagos and Warri, Nigeria.

At 0248 on 10 March *Timor Stream* departed Manzanillo in the Dominican Republic, its final Caribbean port call, with a cargo of refrigerated bananas and a deck cargo of containers bound for Portsmouth in the United Kingdom (UK). Immediately after departure the crew carried out a stowaway search. This was completed at 0325 and the chief and second officers, who had both previously been busy with cargo operations, went to the bridge and reported their findings to the master. The bridge watchkeeping system was in transition from a routine for the vessel's frequent Caribbean port calls to one more suited for the long ocean passage. The master decided that he was best placed to take the bridge watch. He released the two officers to go to their beds and remained alone on the bridge. A lookout was available, but the master decided not to call him.

*Seagate's* chief officer, assisted by an Ordinary Seaman (OS) as lookout, took over the bridge watch on his vessel at 0400. It was reported that *Seagate* was on an auto-pilot heading of 104° at a speed of 10.8 knots.

At 0416 *Timor Stream's* master set an auto-pilot heading of 043° and a speed of 19.5 knots for the 3,592nm track across the Caribbean Sea and Atlantic Ocean (**Figure 1**). At 0441, the master sent a standard departure email (**Annex A**) to the ship's managers and other interested parties from a computer located on the starboard side of the bridge (**Figure 2**). The master noted the ship's position from the Global Positioning System (GPS) at 0500. He wrote it in the deck logbook and plotted the ship's position on the chart; he then adjusted the ship's heading to 041°. Information from the vessel's voyage data recorder (VDR) indicated that the master then occupied himself on the bridge with other tasks.

At around 0515, *Seagate's* lookout alerted the chief officer to a vessel on his starboard side. *Seagate's* transmitted Automatic Identification System (AIS) data showed that she was on a heading of 114° at that time. The chief officer determined that the vessel (*Timor Stream*) was overtaking *Seagate* on a course of around 090° and would pass 3 or 4 cables clear down his vessel's starboard side. He attempted to plot *Timor Stream's* radar target, but was unsuccessful: he did not take a visual bearing of the vessel.

At 0520, the motor yacht *Battered Bull* was heading 326° at a speed of 12.5 knots at a range of 13nm from *Seagate*, on an almost reciprocal heading (**Figure 3**). *Timor Stream* was around 25° on *Battered Bull's* port bow crossing from port to starboard. *Battered Bull's* chief officer identified that *Seagate* and *Timor Stream* were 6.7nm apart and that action was required by *Seagate*, the give way vessel, to avoid a close quarters situation or collision.

By 0532, *Seagate* and *Timor Stream* were 2.8nm from each other on a collision course (**Figure 4**); *Battered Bull* was 8.5nm from *Seagate*. *Battered Bull's* chief officer altered course 24° to port to avoid the developing situation with both *Seagate* and *Timor Stream* (**Figure 5**).

At around 0535, *Seagate's* lookout again told the chief officer about the other ship on his starboard side. As the distance between the two ships further reduced, the lookout yet again alerted the chief officer to the presence of the other ship. At 0539 (**Figure 6**) the chief officer started to alter *Seagate's* heading slowly to port; he predicted that this adjustment would increase the passing distance of the two ships. Shortly afterwards the lookout shouted at the chief officer to "*do something*". The chief officer reported that he saw *Timor Stream* turn to port towards *Seagate* and, when he realised that a collision was imminent, he put the steering controls into manual mode and turned the helm hard to port.

*Timor Stream's* master first saw a ship very close on his port bow at 0540 (**Figure 7**). He was unable to take avoiding action prior to the collision in the short time available to him.

### 1.2.2 The collision

At 0540:50 the two vessels collided (**Figure 8**). *Timor Stream's* heading had not changed; *Seagate's* heading had altered by 7°. *Timor Stream's* bow hit the after part of *Seagate's* starboard side in the area of the accommodation block and engine room; the engine room started to flood and the electrical power system failed. *Seagate's* starboard liferaft was destroyed and its starboard lifeboat fell onto *Timor Stream's* damaged forecastle deck.

### 1.2.3 Actions following the collision

*Seagate's* master went to the bridge. The chief officer activated the ship's general emergency alarm, but it did not sound. As the engine room flooded, the duty engineer and the motorman evacuated it. The steering gear compartment subsequently flooded through the two fire doors that separated it from the engine room.

At 0542 *Timor Stream's* master sounded the general alarm on his vessel. This was followed at 0544 by *Seagate's* master issuing a "Mayday" call by Very High Frequency (VHF) radio. At 0547 *Timor Stream's* master also issued a VHF radio "Mayday" call; *Battered Bull* altered course towards the two damaged ships at 0548.

Shortly after the collision several of *Timor Stream's* crewmen arrived on the forecastle; they shouted to *Seagate's* crew to come across to their vessel. At around 0600, six of *Seagate's* crew climbed on board *Timor Stream*.

Also at around 0600, *Seagate's* master instructed the chief officer to prepare the port side lifeboat and liferaft for launching. The chief officer took a motorman, and told him to lower the lifeboat into the water. The chief officer, who was not wearing a lifejacket, then climbed onto the top of the lifeboat (**Figure 9**), went inside it and attempted to disengage the lifeboat's fall blocks from their release hooks. The chief officer was not familiar with the release mechanism and was unable to release the hooks.

At 0607, full astern power was used on *Timor Stream* to separate her from *Seagate*. No discussion had taken place between the masters of the two vessels before the manoeuvre. *Timor Stream's* crew prepared a pilot ladder on each side of the ship and prepared to launch their rescue boat.



**Figure 9:** Pre-accident photograph showing *Seagate's* starboard lifeboat for illustration purposes

At around 0625 *Seagate's* port liferaft was thrown overboard, and inflated. Eleven crewmen lowered themselves overboard and dropped into the liferaft. The master, chief engineer and second officer remained on board.

*Timor Stream's* master manoeuvred the vessel to facilitate recovery of *Seagate's* crewmen from the liferaft. At 0630 *Seagate's* liferaft was positioned alongside *Timor Stream* and all the crewmen climbed up the pilot ladder and on board.

At about 0630, *Seagate's* chief officer climbed onto the bow of the port lifeboat and attempted to disengage the lifeboat's release hooks manually. Shortly afterwards, at around 0635, he fell overboard from the lifeboat, landing in the sea between the lifeboat and the ship's side. *Seagate's* master used the VHF radio to broadcast that a man was overboard, and the chief engineer threw a lifebuoy into the sea close to the chief officer. The chief officer held onto the lifebuoy and the liferaft boarding ladder. *Battered Bull's* master manoeuvred his yacht (**Figure 10**) close to the chief officer and a manoverboard recovery net was rigged over the yacht's side (**Figure 11**). The chief officer swam, with the lifebuoy, to *Battered Bull* and their crewmen pulled him on board. The chief officer was covered in oil, exhausted and disorientated, but was uninjured.

At 0715 *Seagate's* master broadcast on VHF radio channel 16 that, despite sustaining substantial damage, the vessel was not sinking or in immediate danger.



*Battered Bull's* master contacted *Seagate's* managers in London at 0740 and *Seagate's* chief officer told them about the situation. At 0803, *Battered Bull's* master spoke to the United States Coast Guard (USCG) station in Miami, USA, by satellite telephone. At 0910 the coastguard advised *Battered Bull's* master that a helicopter and fixed wing aircraft would be on site in 10 minutes' time, and that a cutter would arrive in around 2 hours' time. The USCG helicopter and fixed wing aircraft arrived on scene at 0930; the cutter, *USCG Venturous* (Figure 12), arrived at 0958.



**Figure 12:** *USCG Venturous*

At 1015, *Seagate's* chief officer transferred from *Battered Bull* onto *USCG Venturous*. The USCG released *Battered Bull* from the scene at 1017 and it continued on its voyage.

Seventeen crewmen from *Seagate* transferred from *Timor Stream* to *USCG Venturous* at 1100; at 1106 the USCG vessel's commanding officer released *Timor Stream* from the scene to return to port. *Timor Stream's* master headed his vessel for Santo Domingo Bay, to the south of the Dominican Republic.

A boarding team from *USCG Venturous* embarked on *Seagate* at 1215 to assess the vessel's condition. The boarding team's leader handed the master an email that contained a number of instructions from *Seagate's* managers; one of these was to download the data relating to the accident from the ship's VDR. The master interpreted the company's instruction to download the VDR literally. Rather than attempt to save the data or safeguard it by removing the data storage capsule, he left the VDR intact until there was electrical power to allow him to download the data. As a consequence, the VDR's data was not saved or otherwise protected. At 1340 *Seagate's* last three crewmen, including the master, transferred to *USCG Venturous*.

#### **1.2.4 Subsequent actions**

The following day, 11 March 2012, the USCG escorted the master and several officers back on board *Seagate* to stabilise the ship's condition and consider how the vessel might be salvaged. While they were on board, the crew restored some electrical power. Unknown to the crew, this allowed the VDR to restart and the data

## 1.7 REGULATIONS FOR COLLISION PREVENTION

The COLREGS relevant to this accident are quoted in full at **Annex H**. The effects of these COLREGS are summarised as follows:

- Rule 2 – Responsibility. This rule permits a departure from the collision prevention rules if required to avoid immediate danger.
- Rule 5 – Lookout. This rule states that, as well as visual lookout, radar and any other means can, and should, be used when required to assess the risk of collision.
- Rule 7 – Risk of Collision. This rule requires that all means possible, including radar, should be used to assess if a risk of collision exists as early as possible. Risk of collision is primarily determined by monitoring the compass bearing of an approaching vessel.
- Rule 8 – Action to Avoid Collision. This rule requires that action to avoid collision is positive, clear, and made in ample time.
- Rule 13 – Overtaking. This rule states that the overtaking vessel must keep out of the way of the vessel being overtaken. A vessel is defined as overtaking when it is approaching another vessel from a direction more than 22.5° (2 points) abaft its beam. Only the stern light of a vessel being overtaken would be visible from an overtaking vessel.
- Rule 15 – Crossing situation. When two power-driven vessels are crossing each other and there is a risk of collision, the vessel which has the other on its own starboard side shall keep out of the way of the other and, if possible, avoid crossing ahead of the other vessel.
- Rule 16 – Action by the give-way vessel. Every vessel required to give way must take early and substantial action to keep well clear.
- Rule 17 – Action by the stand-on vessel. Where one of two vessels is to keep out of the way the other vessel should maintain its course and speed. The stand-on vessel may take action to avoid collision as soon as it is apparent that the give way vessel is not taking the required actions. If the two vessels are so close, that both vessels would need to take action to avoid collision, then the stand on vessel shall take the best action available to avoid the collision. However, a stand on vessel should try to avoid altering course to port for a vessel on its own port side.

## 1.8 ISM CODE REQUIREMENTS

The ISM Code<sup>3</sup> states inter alia:

- Section 6, Resources and personnel
  - *“The Company should establish procedures to ensure that new personnel and personnel transferred to new assignments related to safety and*

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<sup>3</sup> The IMO International Safety Management Code – Resolution A.741 (18) as amended by MSC.104(730), MSC.179(79), MSC.195(80) and MSC.273(85)

*protection of the environment are given proper familiarization with their duties. Instructions which are essential to be provided prior to sailing should be identified, documented and given.”*

- Section 8, Emergency preparedness
  - “The Company should identify potential emergency shipboard situations, and establish procedures to respond to them.”
  - “The Company should establish programmes for drills and exercises to prepare for emergency actions.”

## 1.9 **BATTERED BULL’S ACTIONS TO AVOID COLLISION**

The privately owned motor yacht *Battered Bull* was on passage from Gustavia, Saint Barthélemy (St. Barts) to West Palm Beach, Florida, USA. At the time of the accident, the bridge was manned by a chief officer and a lookout. The chief officer held a yachtmaster qualification which was suitable for ocean passages on privately owned leisure vessels of up to 200 gross tons (GT). The chief officer and the lookout both saw *Seagate* and *Timor Stream* on their radar and by eye. The two ships were tracked by the radar’s ARPA and displayed, along with their AIS tracks on the yacht’s Electronic Chart System (ECS) (**Figure 24**). The ECS stored the displayed data in its system’s memory.

*Battered Bull*’s chief officer realised, at a range of at least 10nm, that a close quarters situation was developing between his vessel and *Seagate*; he also realised that an alteration of course to starboard would result in a subsequent close quarters situation with *Timor Stream*. The chief officer made an early and bold alteration of course to port to avoid any risk of collision with the other two ships.

## 1.10 **PREVIOUS/SIMILAR ACCIDENTS**

Three major collisions of SOLAS sized vessels<sup>4</sup> were reported to the MAIB at around the same time as this accident and were investigated by the Branch. In all three accidents, the actions of the watchkeepers were significant causal factors and in all three cases the key players were senior navigating officers who should have been fully familiar with their responsibilities as watchkeepers:

- In December 2011, the Panamanian registered container ship *ACX Hibiscus* and the UK registered container vessel *Hyundai Discovery* collided in the eastern approaches to the Singapore Strait.
- In March 2012, the UK registered passenger ferry *Stena Feronia* and the Cook Islands registered general cargo vessel *Union Moon* collided in the approaches to Belfast Harbour.
- In March 2012, the Dutch registered reefer container vessel *Spring Bok* and the Maltese registered liquefied petroleum gas (LPG) carrier *Gas Arctic* collided 6nm south of Dungeness.

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<sup>4</sup> Greater than 500 Gross Tons (GT)

## SECTION 2 – ANALYSIS

### 2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

### 2.2 THE COLLISION

*Seagate* was well established on a long sea passage; *Timor Stream* had left port 3 hours earlier at the beginning of the voyage to the UK. *Seagate*'s chief officer saw *Timor Stream*, but assumed that he was being overtaken and that *Timor Stream* would pass clear of *Seagate*. *Timor Stream*'s master, on watch alone, was not keeping an effective lookout. Neither watchkeeper was aware that the two ships were on a collision course.

*Seagate*'s chief officer realised that a collision was imminent around 1 minute before it occurred. He turned his vessel's helm full to port for a few seconds; however, this was too late to have any significant effect. *Timor Stream*'s master was not aware of *Seagate*'s position until moments before the collision and had no time to take avoiding action.

### 2.3 SEAGATE – ACTIONS LEADING TO THE COLLISION

#### 2.3.1 Effective lookout

*Seagate*'s OS lookout repeatedly alerted the chief officer to the presence of a vessel approaching from the starboard side. Having decided that the reported vessel was overtaking, the chief officer dismissed the lookout's reports without making adequate checks himself. Had he done so, he would have established that *Timor Stream* was on a steady bearing, and that his was the give-way vessel. His actions were contrary to the COLREGS' Rule 5 (lookout) which requires watchkeepers to assess the situation, and the risk of collision.

The chief officer's inaction showed a total disregard for the safety of his vessel and his shipmates. Despite his experience as an OOW and gaining the required level of knowledge of the COLREGS, he made an unfounded assumption that *Timor Stream* would pass clear of his vessel.

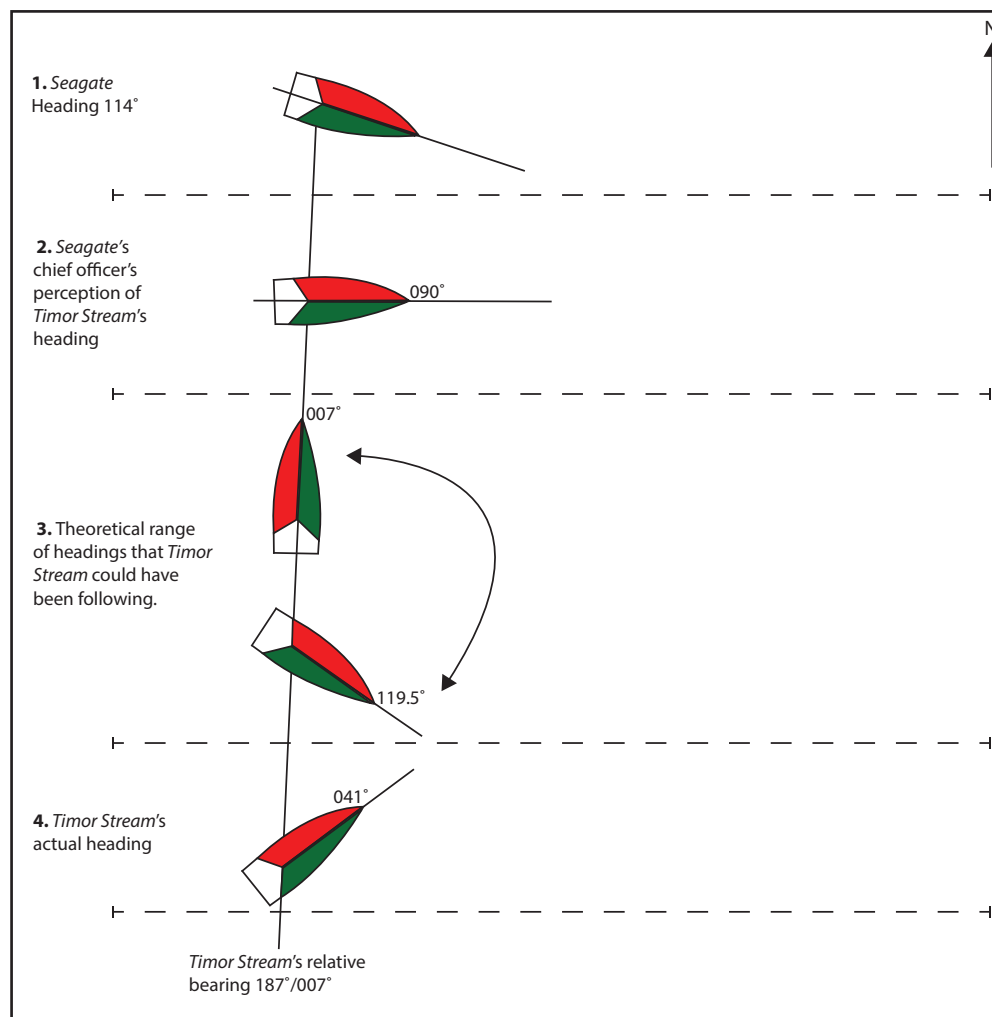
#### 2.3.2 Assessment of the situation

*Seagate*'s chief officer believed that *Timor Stream* was on a heading of around 090°, at a speed capable of overtaking *Seagate* so that it would pass his vessel with a CPA of 3 or 4 cables. *Battered Bull*'s ECS data showed that, for the 40 minutes before the collision, *Timor Stream* was on a constant heading of 041° at a speed of 19.5 knots. *Timor Stream* was on a steady compass bearing of 187° from *Seagate*, 17° forward of *Seagate*'s starboard beam (**Figure 25**).

*Timor Stream* displayed the appropriate navigation lights for a ship of its size when underway. Its two masthead lights and port-side light could have, in theory, shown *Seagate*'s chief officer that *Timor Stream*'s heading was between 007° and 119.5° (**Figure 26**). Had the chief officer established *Timor Stream*'s actual heading using



his radar's ARPA, he should have identified that his estimate of *Timor Stream*'s heading differed by about  $50^\circ$  from its actual heading; he should also have realised that *Seagate* was in a crossing situation rather than an overtaking one.



**Figure 26:** Analysis of the potential range of headings being followed by *Timor Stream* based on viewing its navigation lights alone

When *Seagate*'s lookout first reported the presence of *Timor Stream* to the chief officer, it was forward of the starboard beam. *Timor Stream* remained forward of *Seagate*'s beam throughout, on a steady bearing of  $187^\circ$  and with a steady aspect, until the two ships collided. The chief officer did not receive any indication that the relative bearing of the other vessel was changing to support his assumption that *Timor Stream* was overtaking. Had he given any thought to what was being reported to him, it should have prompted him to consider that further assessment was required.

If the chief officer had himself taken steps to verify the lookout's reports, for example by taking a series of compass bearings of the ship or plotting it using his radar's ARPA, he would have identified that *Timor Stream* was on a steady bearing. He would then have had ample time to establish that his was the give-way vessel, and been able to take appropriate action to avoid the collision.

The chief officer did not assess the situation correctly initially, or reconsider his assessment when the situation did not develop as he expected. Even in the scenario that he had imagined, he was willing to accept a CPA that was much less than specified in the master's night orders. Unfortunately, he was not able to explain why his actions had fallen so far short of what was normally expected of him.

### **2.3.3 The chief officer's perception of *Timor Stream*'s aspect**

The chief officer estimated from the aspect of *Timor Stream*'s navigation lights, that the vessel was on a heading of around 090°. The vessel's actual heading was 041°, a discrepancy of almost 50° (**Figure 26**). A few seconds before the collision, the chief officer believed that *Timor Stream* turned to port, although the vessel's heading did not change at all. It is considered likely that the 'turn' reported by the chief officer was actually him correcting his false perception of *Timor Stream*'s aspect.

This accident emphasises the importance of maintaining a methodical approach to watchkeeping and collision avoidance. It is extremely difficult to determine a vessel's aspect at night and, even if correct, aspect is no guarantee of a vessel's actual heading or course.

### **2.3.4 Human factors**

*Seagate*'s chief officer was qualified to be the master of deep sea ships of unlimited size and had been a bridge watchkeeper for many years. However, on the morning of the accident he felt able to ignore the basic watchkeeping requirements of keeping an effective lookout and establishing whether a risk of collision existed with an approaching ship. Specifically, he had a misplaced confidence in his ability to determine by eye alone whether a risk of collision existed, and felt able to ignore the risks of making a misjudgment based on scanty information.

The chief officer's attitude could have been influenced by several factors. Travelling at 11 knots, *Seagate* was a relatively slow ship and the chief officer could have expected that faster ships would routinely overtake his vessel. He might also have been lulled into a false sense of security by the conditions: it was a warm, occasionally moonlit night, with light to moderate winds, good visibility and little other traffic in an open sea. In these circumstances, it is possible that an otherwise experienced watchkeeper might allow himself to underestimate the risks of a collision.

The chief officer's complacent attitude to the required watchkeeping standards, his misplaced belief in his ability to assess the risk of collision by using visual observation, and his underestimation of the chance of encountering another vessel at close quarters combined to prevent him from taking the necessary actions to prevent the collision.

### **2.3.5 Fatigue**

There was no evidence that *Seagate*'s chief officer was fatigued at the time of the accident, and fatigue was not considered to have had any effect on his actions.

## SECTION 3 – CONCLUSIONS

### 3.1 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE BEEN ADDRESSED OR HAVE NOT RESULTED IN RECOMMENDATIONS

#### *Seagate*

1. *Seagate's* chief officer did not establish that *Timor Stream* was on a steady bearing, and that his was the give-way vessel. His actions were contrary to the COLREGS' Rule 5 (lookout) which requires watchkeepers to assess the situation, and the risk of collision. [2.3.1]
2. The chief officer's inaction showed a total disregard for the safety of his vessel and his shipmates. Despite his experience and knowledge of the COLREGS, he made an unfounded assumption that *Timor Stream* would pass clear of his vessel. [2.3.1]
3. The chief officer did not assess the situation correctly initially, or reconsider his assessment when the situation did not develop as he expected. Even in the scenario that he had imagined, he was willing to accept a CPA that was much less than specified in the master's night orders. [2.3.2]
4. It is considered likely that the 'turn' seen by the chief officer was actually him correcting his false perception of *Timor Stream's* aspect. This accident emphasises the importance of maintaining a methodical approach to watchkeeping and collision avoidance. It is extremely difficult to determine a vessel's aspect at night and, even if correct, aspect is no guarantee of a vessel's acting heading or course. [2.3.3]
5. The chief officer's complacent attitude to the required watchkeeping standards, his misplaced belief in his ability to assess the risk of collision by eye, and his underestimation of the chance of encountering another vessel at close quarters combined to prevent him from taking the necessary actions to prevent the collision. [2.3.4]
6. The chief officer was unable to operate the lifeboat release mechanism because he did not know how it worked. [2.8.1]
7. Had the chief officer's fall overboard gone unnoticed, or had he been injured as he fell between the lifeboat and the ship, he could have easily drowned. By not wearing a lifejacket while attempting such a risky task the chief officer placed himself in unnecessary danger and required others to come to his rescue. [2.8.2]
8. The chief officer had only been on board for 10 days prior to the accident. However, his rank placed him in a position where he would be expected to take a leading role in responding to any emergency on board and, as required by the ISM code, he should therefore have been familiar with the operation of all the lifesaving appliances and emergency procedures on board. [2.8.3]
9. A more detailed evaluation system would have helped *Zodiac's* managers improve their knowledge of crew performance. There are also benefits in reporting officers routinely providing more immediate feedback to crew on their performance. *Zodiac's* managers have since implemented a more open and wide ranging form of evaluation which should be more beneficial to them and their personnel. [2.9.2]

### ***Timor Stream***

1. *Timor Stream*'s master decided that it was better to let the nominated night watchkeepers go to bed and for him to take the watch himself. However, the master did not post a lookout or set the watch alarm, and instead relied on his ability to maintain an effective lookout on his own. [2.4.1]
2. The master allowed himself to be distracted by tasks other than keeping a lookout, possibly by placing himself in a position where he could not see out of the bridge windows or look at his navigation aids. In choosing to take the watch alone and not setting the watch alarm the master demonstrated extremely poor judgment, systematically overcoming each of the safeguards that should have been in place for keeping an effective navigational watch. [2.4.2]
3. The master set the standards of watchkeeping on board, yet he did not meet the minimum standards that he would have expected of his officers. His lack of regard for his primary roles as lookout and officer in charge of a navigational watch can only be assessed as complacent. [2.4.3]

### ***Battered Bull***

1. *Battered Bull*'s chief officer showed a thorough understanding of the navigational situation that he faced, and took the appropriate professional actions expected of a diligent watchkeeper leading an effective bridge team. [2.5]
2. The skilful and highly responsible actions of *Battered Bull*'s master, chief officer and crewmen in their response to the collision are highly commendable. [2.7.2]

### **Other conclusions**

1. This, and the other accidents recently reported to the MAIB, should serve as powerful reminders to officers in charge of a navigational watch, and their employers, the importance of maintaining high standards of watchkeeping at all times. [2.6]
2. Masters and bridge watchkeepers must think carefully before attempting to manoeuvre their vessels after a major accident. The extent and effect of any damage must be considered carefully to make sure that any manoeuvres do not make the situation more hazardous. [2.7.1]
3. The nature of both *Seagate*'s and *Timor Stream*'s trading patterns made it difficult for managers to conduct sea-going audits or personnel assessments. This accident and the other similar accidents show that there is a need for company managers to implement systems which seek to identify and correct poor professional standards. [2.9.1]
4. VDR data is a significant aid to accident investigation, and the data is equally beneficial to ship managers, masters and their crew. In the accident, the total lack of VDR data from one vessel, and incomplete data from the other significantly hampered the reconstruction of events. Ship managers should ensure that the VDRs on their vessels can both record and be downloaded correctly. One way, and perhaps the best way, to have confidence that a vessel's VDR will work when it is needed is to practise recording, downloading and interrogating the data regularly. [2.10]